MODULE 4 · THREATS AND HUMAN FACTORS

Lesson 15: CRM, Monitoring and Intervention

Lesson 15 of 24 · about 8 min

LEARNING OBJECTIVES

- Describe the role of the pilot monitoring in upset prevention.
- Use standard callouts and intervention techniques.
- Apply clear transfer of control during an upset.



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Many developing upsets are first visible to the pilot who is **not** flying. Effective monitoring means actively checking the flight path, the automation modes and the energy state — not just looking at the instruments.

- Monitor during periods of low workload as carefully as during high workload. Developing upsets often start in cruise.
- Watch for trends: speed decaying, bank slowly increasing, pitch rising, vertical speed growing.
- Plan monitoring during high-workload phases, such as approach and go-around, so that someone is always watching the flight path.

Callouts

Use the standard callouts of your Operations Manual. Typical examples:

Deviation	Typical callout
Speed below target or trend towards the low-speed band	“Speed”

Excessive bank	“Bank angle”
Excessive pitch	“Pitch”
Excessive descent rate	“Sink rate”
Upset recognised	“Nose high” / “Nose low”, with the bank direction

The pilot flying acknowledges and corrects. If there is no response, the pilot monitoring repeats the callout, then states the action needed — and is ready to take control.

Intervention and transfer of control

- If the pilot flying does not respond to callouts, or is clearly incapacitated or disoriented, the pilot monitoring **takes control**: “I have control.”
- Transfer of control must be **clear and acknowledged** — “You have control.” Never let two pilots make inputs at the same time; on side-stick aircraft, dual inputs are summed.
- The company supports any crew member who intervenes in good faith to protect the flight.

Communication in a recovery

- The pilot flying announces the upset and the recovery: “Nose low, rolling left, recovering.”
- The pilot monitoring calls attitude, speed and altitude, watches for g limits and the stall warning, and reads relevant checklists only once the aircraft is stable.
- Inform ATC as soon as possible after control is regained.

KEY TAKEAWAYS

- The pilot monitoring is often first to see a developing upset — monitor actively, especially trends.
- Use standard callouts: speed, bank angle, pitch, sink rate, nose high / nose low.
- Transfer control clearly: “I have control” — “You have control.” No dual inputs.

KNOWLEDGE CHECK

The pilot flying does not respond to two “Bank angle” callouts as bank increases through 50°. What should the pilot monitoring do?

A. Wait for the autopilot to correct

B. Take control clearly with “I have control” and recover

C. Make an input on the controls without saying anything

D. Inform ATC first

Answer: B. If callouts are not answered, the pilot monitoring takes control with a clear, acknowledged transfer, then recovers the aircraft.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.